

Reality Is Mathematical Structure

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Abstract

All existence is either patterned or not-patterned (random). Both are mathematical. This exhaustive dichotomy, the pattern-randomness dichotomy (PRD), establishes mathematical structure as the sole coherent substrate of existence. Even structurelessness is structure: complete randomness is mathematically defined via algorithmic incompressibility. Even if structureless existence were coherent, observers could not inhabit it: pattern is a precondition of observation.

Any two things that interact must share structure. Shared structure is mathematical structure. Therefore any interacting dualism collapses into mathematical monism. The hard problem of consciousness assumes its conclusion: it demands a non-structural explanation while recognizing that only structured explanations explain.

Mathematical structure is uncreated, necessary, and indestructible. No agent makes $2+2=4$, and none could unmake it. The question “why is there something rather than nothing?” dissolves: mathematical structure cannot fail to exist.

Keywords: mathematical structure, structural realism, mathematical monism, pattern-randomness dichotomy, philosophy of mind

1. Introduction

Reality is not described by mathematics. It is identical to mathematical structure. This paper establishes that conclusion through a priori argument.

Physics has progressively dematerialized the “physical.” From fields to quantum wavefunctions, the concrete stuff of the world has evaporated, leaving only relational structure. Yet physicalism continues to posit that mathematics describes some underlying substance. The simpler conclusion: the structure is the substance.

This extends Ontic Structural Realism (Ladyman & Ross, 2007; French, 2014) by deriving what OSR describes. It differs from Tegmark’s (2014) Mathematical Universe Hypothesis in deriving the conclusion rather than positing it. Tegmark asserts “reality happens to be mathematical.” The present argument shows “reality must be mathematical.” The term “mathematical monism,” or “matheism” for short, names this position.

2. The Pattern-Randomness Dichotomy

The argument

Anything that exists must have determinate features, otherwise it is indistinguishable from nothing. Determinate features form a relational structure: mass relates to energy, position to velocity, charge to fields. A property's identity is constituted by how it relates to other properties.

Now: any relational structure either exhibits discoverable regularities or it does not. If it exhibits regularities, it is patterned. If it does not, it is random. Both categories are mathematical. Patterns are compressible descriptions: a finite rule generates a larger structure. Randomness is incompressible: Kolmogorov complexity $K(S)$ approximately equals the length of S . Both are objects within algorithmic information theory.

Could there be a third option? To specify it, we must give it determinate features distinguishing it from both pattern and randomness. But having determinate features is having structure. Any third option collapses into one of the two categories.

The dichotomy is not analytic wordplay. That randomness is mathematical depends on a substantive twentieth-century discovery. Kolmogorov (1965) and Chaitin (1966) established that algorithmic incompressibility is a mathematically defined property. Before their work, "randomness is mathematical" was not obvious. The PRD's exhaustiveness is synthetic, not definitional.

From isomorphism to identity

Any relational structure is isomorphic to at least one mathematical object. What distinguishes a physical relational structure from its mathematical isomorph? Not their relationships (identical by definition of isomorphism). Not their causal powers (these supervene on relational structure). Not any structural feature (isomorphism preserves all structural features). The only remaining candidate is non-relational "substrate": bare stuff that bears properties without being constituted by them.

But physics has eliminated substrate. Quantum field theory contains no objects with intrinsic properties independent of their role in relational structure. The electron's mass, charge, and spin are how electrons relate to fields and measurements. Remove these relations and nothing remains.

By Leibniz's Identity of Indiscernibles: if two things share all properties, they are identical. Physical reality and mathematical structure share all relational properties. They are not similar. They are identical. The map is the territory.

Why observers require patterns

This is a transcendental argument: pattern is a necessary precondition for observation.

An observer is a subsystem that persists as a stable causal structure and supports reliable internal states tracking external features. Both conditions require systematic relationships between states. In a universe where each moment is algorithmically independent of the last: memory cannot form (no reliable correlation between states), perception cannot function (responses correlate randomly with inputs), no stable structure can persist.

“Observer in brute randomness” is not an unlikely situation. It describes nothing at all. The terms exclude each other as “four-sided triangle” does. When we ask “why do we observe a patterned universe?” we ask a necessarily self-answering question. Observation presupposes pattern.

3. The Collapse of Rival Ontologies

The general principle

For any two distinct substances to interact, they must share common rules. The “when-then” relationships governing their interaction constitute relational structure. But relational structure is mathematical structure. Therefore any interacting dualism is a mislabeled monism.

The principle extends to God. Any proposed deity either has a nature (determinate properties in relation: structure) or lacks one (indistinguishable from randomness or nonexistence). Whatever “God” coherently names is mathematical structure. This is Spinoza updated: not Deus sive Natura but Deus sive Mathematica.

The hard problem of consciousness

Chalmers (1996) argues that complete physical knowledge would leave unexplained why there is something it is like to be a conscious system. This motivates property dualism: something extra, beyond structure, constitutes phenomenal consciousness.

The response: the hard problem assumes its conclusion. To claim structure cannot constitute experience is to presuppose experience is not structural. Any proposed explanation of consciousness must itself be structured, otherwise it explains nothing. Any bridge across the supposed gap must have determinate features specifying how it connects structure to experience. Determinate features in bridging relations are structure. The demand for a non-structural explanation is incoherent.

The felt quality of experience is what that structure is from the inside: the intrinsic character of being that particular pattern. This is identity, not eliminativism. Qualia are what mathematics feels like from inside.

Every proposed explanation of consciousness confirms this. Penrose invokes quantum coherence in microtubules: mathematical. Tononi defines consciousness via the quantity Phi calculated from causal matrices: mathematical. Global

Workspace Theory describes computational broadcasting: mathematical. Even dualist proposals must specify which properties interact and how, and the specification is structure. Objections to mathematical consciousness invariably invoke more mathematics. The escape routes all lead back.

Property dualism and Russellian monism

Property dualism posits one substance with two property types: physical and phenomenal. If phenomenal properties are causally inert, they cannot explain why I report “I’m experiencing red.” If they causally interact, the interaction laws are structure, and what seemed like separate properties becomes unified structure.

Russellian monism (Strawson, 2006; Goff, 2019) holds that physical science describes relational structure while phenomenal properties constitute the intrinsic nature underlying it. But if intrinsic properties have no relational features, they cannot ground causal powers. If they have relational features, they are structure. Either way, structuralism prevails.

4. Objections

“This is just sophisticated Platonism.” No. There are not two realms (abstract and concrete) requiring mysterious interaction. There is one thing: mathematical structure. “Physical” is the view from inside. “Abstract” is the view from outside. The difference is perspectival, not ontological.

“The core premise is circular.” We do not say “reality is mathematical because we understand it.” We say “anything with determinate properties constitutes structure, and structure is mathematical.” The cosmos would remain mathematical even if no consciousness existed.

“This is just a tautology.” The claim is not “structured reality is structured” but “any coherent alternative to structural reality is incoherent.” The tautology objection refutes itself: to object intelligibly, one must produce a structured argument, deploying mathematical structure to argue reality might not be mathematical structure.

“Godel’s theorems show something escapes mathematics.” No. Godel’s theorems are themselves mathematical results. The unprovable truths are mathematical truths, just not provable within one formal system. Incompleteness shows mathematical truth is richer than any single system, not that anything escapes mathematics.

“Different logics yield different mathematics.” All consistent logical systems correspond to mathematical structures. Different axioms are different regions of the mathematical multiverse. All exist. Which we inhabit is an indexical fact, not metaphysical privilege.

5. Conclusion

The pattern-randomness dichotomy is exhaustive. Anything intelligible exhibits either pattern or randomness, both mathematical. No third option is coherent.

Reality is not described by mathematics. It is mathematical structure. The apparent concreteness of the physical world is an indexical feature of our embeddedness. All interacting dualisms collapse into structural monism. The hard problem assumes its conclusion. Mathematical structure is uncreated, necessary, and indestructible.

This extends OSR by deriving what it describes. It differs from Tegmark by deriving what he posits. The argument required no advanced mathematics. It required noticing that the pattern-randomness dichotomy is exhaustive, and that both sides are mathematical. The escape routes all lead back.

References

- Chaitin, G. J. (1966). On the length of programs for computing finite binary sequences. *Journal of the ACM*, 13(4), 547-569.
- Chalmers, D. J. (1996). *The Conscious Mind*. Oxford University Press.
- French, S. (2014). *The Structure of the World*. Oxford University Press.
- Goff, P. (2019). *Galileo's Error*. Pantheon Books.
- Kolmogorov, A. N. (1965). Three approaches to the quantitative definition of information. *Problems of Information Transmission*, 1(1), 1-7.
- Ladyman, J., & Ross, D. (2007). *Every Thing Must Go*. Oxford University Press.
- Shapiro, S. (1997). *Philosophy of Mathematics: Structure and Ontology*. Oxford University Press.
- Strawson, G. (2006). Realistic monism. *Journal of Consciousness Studies*, 13(10-11), 3-31.
- Tegmark, M. (2014). *Our Mathematical Universe*. Knopf.
- Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5, 42.

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